

NN Sheet 4

- 1- What problem could occur in SOM learning if you use a very small neighborhood? Why?
- 2- SOMs can reduce the dimensionality of a given data space. Explain what that means. Please give an example of how this capability can be used for practical applications.
- 3- Why are SOMs interesting for researchers who study biological nervous systems?
- 4- In a 2-D self-organizing map with input vectors of dimension m , and k neurons in the map, how many weights will there be?
- 5- Describe the competitive process of the Self-Organizing Map algorithm.
- 6- In unsupervised learning, there are no target output values. So, what does unsupervised learning learn?
- 7- How many layers are there in a Kohonen network (Self Organizing Map) and what do they do?
- 8- How are the weights in a Kohonen network (Self Organizing Map) updated?
- 9- How can a Kohonen network (Self Organizing Map) be used to compress data with minimum loss of information?
- 10- Describe the architecture of the Self Organizing Map (SOM) known as a Kohonen network.
- 11- What can SOM network be useful for?
- 12- The self-organizing process can be said to have four major components: Initialization, Competition, Cooperation, and Adaptation. Briefly describe how each of these components is implemented.

- 13- Explain what is meant by the term Self Organising Map (SOM)?
- 14- What is Density Matching, and why is it an important feature of SOMs?
- 15- Outline the main components of the self-organising process that is used to train a Kohonen Network.
- 16- Describe the properties of the mapping that is performed by an appropriately trained Kohonen Network.

17- There is an excellent online database of handwritten digits that is well-suited for programming projects:

<http://yann.lecun.com/exdb/mnist/>

It contains a training set of 60,000 digitized 28×28 pixel bitmap images of handwritten digits, and a test set consisting of another 10,000 images. To simplify access and save memory, the images are encoded in a very simple format. All details are provided on the website. As you can see, many different neural networks have been tested on this database, and the resulting misclassification rates are listed.

Apply a Self-Organizing Map (Kohonen Map) to our hand-written digit database. Use the Euclidean distance to determine the winner neuron in the output (map) layer – then you will need only 784 input neurons, since there is no dummy input required. The output layer consists of 10×10 neurons. You can experiment with different neighborhood functions and learning rates and determine the ones that work best. You should definitely reduce the size of the neighborhood and the learning rate from epoch to epoch, though. In order to visualize the state of the output layer before and during learning, implement an integer array of size ten for each of the 100 output neurons. These variables will count how often a given neuron wins for a given digit (0 to 9). First, set all of the counters to zero. Then input each of

the 10,000 test samples into the network once and determine the winner as usual, without updating any weights. For the winner neuron, increase its counter that represents the digit associated with the current input. After this process is finished, determine for each neuron the digit for which the neuron won most often. Now output the resulting 10×10 matrix of digits, which indicates the selectivity (= preferred digit) of each neuron in the map.

(a) As usual, start with random weights and visualize the resulting map as explained above. Then perform one epoch of learning, and visualize the map once again. Do the same after 10 and after 100 epochs. If the map still looks random after 100 epochs, try modifying your neighborhood function and/or learning rate and their change between epochs, or try running more epochs, until you can see some learning effect. Submit all four resulting maps either as files or hardcopies.

(b) Run your network again twice with the maximum number of epochs that you used in (a), but of course each time with different initial weights and a different random order of exemplars. Submit the resulting two maps. Compare the three final maps (the one from (a) and the two maps from (b)). What is similar in all of them, and what is different? Explain why.